Evidence for Polygenic Adaptation to Pathogens in the Human Genome

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Associate editor: John Novembre

Abstract

Most approaches aiming at finding genes involved in adaptive events have focused on the detection of outlier loci, which resulted in the discovery of individually "significant" genes with strong effects. However, a collection of small effect mutations could have a large effect on a given biological pathway that includes many genes, and such a polygenic mode of adaptation has not been systematically investigated in humans. We propose here to evidence polygenic selection by detecting signals of adaptation at the pathway or gene set level instead of analyzing single independent genes. Using a gene-set enrichment test to identify genome-wide signals of adaptation among human populations, we find that most pathways globally enriched for signals of positive selection are either directly or indirectly involved in immune response. We also find evidence for long-distance genotypic linkage disequilibrium, suggesting functional epistatic interactions between members of the same pathway. Our results show that past interactions with pathogens have elicited widespread and coordinated genomic responses, and suggest that adaptation to pathogens can be considered as a primary example of polygenic selection.

Key words: **human evolution, pathway analysis, adaptation, polygenic selection, epistasis.**

Introduction

Since the emergence of modern humans in Africa (Clark et al. 2003; McDougall et al. 2005) and their migrations into the rest of the world around 50–60 kya, human populations have faced many challenges arising from the colonization of new habitats, such as changes in food sources, pathogen load, and climatic conditions (Balaresque et al. 2007). Adaptation to local environments is expected to have left its signature in the human genome, but identifying loci involved in such adaptive events has proven to be difficult given that both selection and demographic processes can have confounding effects on the observed patterns of genetic diversity (Nielsen et al. 2007; Excoffier, Foll, et al. 2009).

In the last decade, genome scans for selection have detected multiple signals of genetic adaptations in recent human history (Kayser et al. 2003; Storz et al. 2004; Voight et al. 2006; Wang et al. 2006; Sabeti et al. 2007; Williamson et al. 2007; Barreiro et al. 2008; Fumagalli et al. 2011). In addition to the identification of new genes putatively under selection, they have confirmed selection candidates found earlier in studies targeted on specific phenotypes, such as lactase persistence (Enattah et al. 2002) or skin pigmentation (Izagirre et al. 2006). Many of these genome scans aimed at the detection of strong selective sweeps (with genomic regions of low diversity surrounding beneficial mutations), using tests based on the site frequency spectrum (Williamson et al. 2007), extended haplotype homozygosity (Voight et al. 2006), linkage disequilibrium (LD) (Wang et al. 2006), or population subdivision (Kayser et al. 2003; Barreiro et al. 2008) (as reviewed in Nielsen et al. 2007; Akey 2009). However, adaptation can also result from selection on standing variation (Pritchard et al. 2010), and the approaches described earlier have little power for the detection of such "soft sweeps." An alternative method is to correlate changes in allele frequencies with environmental parameters, which has been applied successfully to find that additional variables such as climate (Young et al. 2005), diet (Hancock, Witonsky, et al. 2010), and pathogens (Fumagalli et al. 2011) were involved in human adaptations.

Still, most of the studies aiming at identifying selection are based on the detection of single outlier loci, whereas genome-wide association studies (GWAS) have revealed that many traits are affected by multiple loci, each contributing modestly to the phenotype (Stranger et al. 2011), possibly through epistatic interactions (Phillips 2008). It is thus likely that adaptive events acting upon such polygenic traits arose from standing variation rather than from new mutations, and that they resulted in small changes in allele frequency

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at several loci (Pritchard et al. 2010). To detect this more subtle form of selection, we propose a gene set enrichment approach where we jointly analyze data from many loci to gain insight in how selection has affected specific pathways. Although a number of pathways have been recognized as being under selection from genome scans (Fumagalli et al. 2011), gene set enrichment methods fundamentally differ from a posteriori testing for Gene Ontology (GO) process enrichment. Rather than testing whether candidate loci (identified as being significant at a given level or just as top outliers) are overrepresented for some GO categories, gene set enrichment approaches test whether the distribution of statistics computed across all genes of a given gene set (e.g., a given biological pathway) statistically differs from genome-wide expectations.

Gene set enrichment approaches have originally been developed for and successfully applied to gene expression studies (Mootha et al. 2003; Sweet-Cordero et al. 2005), as significant associations with phenotypes were often not detectable for individual genes. The general idea is to rank genes by their difference in expression between phenotypes, and then test whether a predefined group of genes (e.g., from a given pathway) is enriched at the top or bottom of this list. This strategy should be biologically meaningful as most biological functions and phenotypes result from a cascade of events in a pathway, or from physical interactions between proteins or metabolites.

One of the first published methods, the gene set enrichment analysis (GSEA) (Subramanian et al. 2005), uses a weighted version of the Kolmogorov–Smirnov test to assess the enrichment score of a pathway. This methodology is still widely used, and several flavors and software implementations of GSEA have been developed since (Subramanian et al. 2007; Wang et al. 2007; Holden et al. 2008; Zhang et al. 2010). Despite its success, GSEA has been shown to perform poorly in comparison with other methods (Kim and Volsky 2005; Dinu et al. 2007; Efron and Tibshirani 2007; Tintle et al. 2008, 2009; Tsai and Chen 2009). For example, a simple parametric test where one takes the sum (SUMSTAT; Tintle et al. 2009) or the mean of the test scores of all genes in a pathway usually gives better results than GSEA. It also performs often as well as other more complex methods (Ackermann and Strimmer 2009). Gene set enrichment methods have been further developed for single nucleotide polymorphism (SNP) data from GWAS (Wang et al. 2007; Holden et al. 2008; Nam et al. 2010) and their application has successfully been used in the investigation of common diseases, revealing pathways containing genes that would individually not show any significant association (Baranzini et al. 2009; Menashe et al. 2010).

The fact that some gene networks could harbor a series of small effect mutations leading to a disease phenotype gives credence to the idea that, reciprocally, several small effect mutations could also be involved in adaptations, leading to globally improved functionalities of a given pathway or protein complex. In this study, we use a gene set enrichment approach to uncover signals of recent adaptive events that may have occurred among human populations. We detect many pathways enriched in signals of selection, but most of them contain genes that are shared among various pathways. After correcting for this overlap, we focus our analysis on the remaining top scoring gene sets, and investigate possible epistatic interactions by testing for long distance genotypic LD.

Results and Discussion

Multiple Pathways Are Enriched for Adaptive Signals

Positive selection acting in one or a few populations should increase global genetic differences between populations. We therefore used the degree of population differentiation measured by F_{ST} and computed over many worldwide populations as a proxy for positive selection at the SNP level. The use of F_{ST} to detect adaptation has a long tradition (reviewed in Beaumont 2005), and it has been shown to be a powerful statistic to evidence recent adaptations (Innan and Kim 2008). Although other statistics have been developed to detect selection from genome scans within single populations (Nielsen et al. 2005; Sabeti et al. 2007; Zhai et al. 2009; Pavlidis et al. 2010), F_{ST} has the advantage of being sensitive to adaptations occurring in different parts of the range of a species and therefore to collect information from various populations into a single statistic. We downloaded the SNP data set of the Human Genome Diversity Panel (HGDP) consisting of 660,918 SNPs genotyped in 53 populations (Cann et al. 2002; Li et al. 2008). After processing the data as described in the Materials and Methods, 660,470 SNPs remained within 51 populations. To assess the significance of the F_{ST} values of these SNPs, we performed simulations based on a hierarchical-island model of population structure (Excoffier, Hofer, et al. 2009) to take into account the fact that some populations share a recent history, which has been shown to fit the observed genomic patterns of human genetic structure much better than a simpler finite island model (Excoffier, Hofer, et al. 2009). F_{ST} probabilities were then transformed into z scores, with extreme positive (respectively negative) values indicating relative high (respectively low) levels of population differentiation. These z scores have been shown to be approximately normally distributed (Hofer et al. 2012).

We tested a total of 1,043 gene sets, as defined in the NCBI Biosystems database (Geer et al. 2010), for enrichment in signals of positive selection using the SUMSTAT approach (Tintle et al. 2009), which takes the sum of a summary statistic associated to the genes in a given gene set. As a summary statistic, we used here the highest z score among SNPs within 50 kb of a given gene (supplementary table S1, Supplementary Material online). We assessed the significance of the observed SUMSTAT scores by comparing their value with scores of random gene sets, while controlling for SNP density. This correction is necessary as genes containing many SNPs are likely to have a larger z score, resulting in the spurious detection of pathways enriched for genes with a high SNP density as being under selection. To correct for this potential bias, we assigned genes to bins according to their SNP density (supplementary table S2 and fig. S1, Supplementary Material online) and standardized their z score based on the distribution of z scores within their bin (see Materials and Methods).

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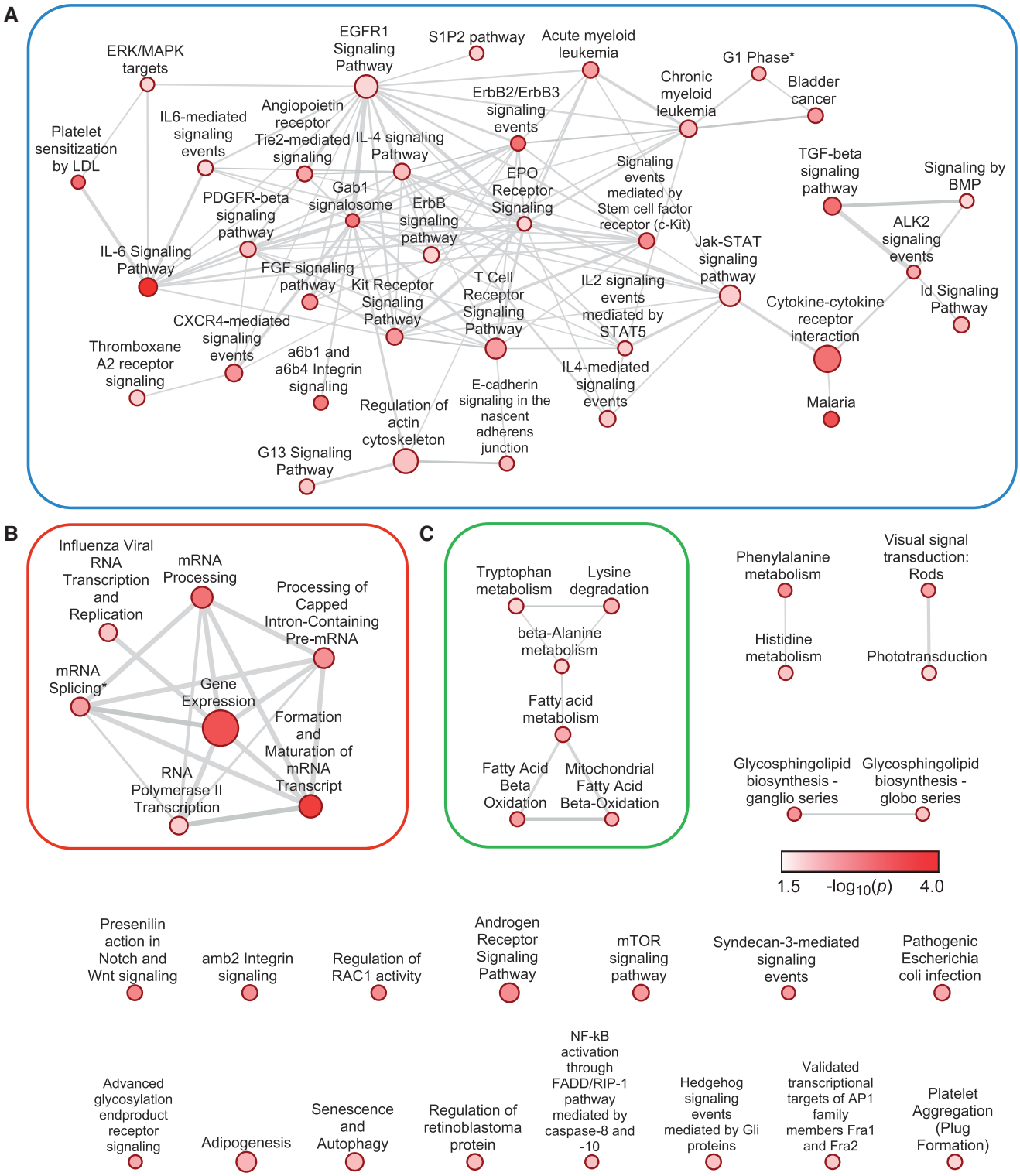


Fig. 1. Gene sets enriched for signals of positive selection. The 70 nodes represent gene sets with q values ≤ 0.2. The size of a node is proportional to the number of genes in a gene set. The node color scale represents gene set P values. Edges represent mutual overlap; nodes are connected if one of the sets has at least 33% of its genes in common with the other gene set. The widths of the edges scale with the similarity between nodes. Rectangles *A*, *B*, and *C* mark the three large clusters of connected gene sets as discussed in the main text. (Nodes marked with * represent unions of pathways that share more than 95% of their genes.)

Among the 1,043 gene sets tested, we found 70 candidate sets with a q value lower than 20% (supplementary table S1, Supplementary Material online), a number that is significantly higher ($P < 0.01$) than genome-wide expectations (i.e., as measured by random permutations of z scores across all genes) (supplementary fig. S2, Supplementary Material online). However, we observed a considerable overlap of genes among the 70 pathways, as shown in figure 1 where

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Table 1. Candidate Pathways for Positive Selection after Removing Overlapping Genes from Less Significant Gene Sets ("pruning").

Rank	Gene Set[a,b]	Set Size before/after Pruning	*P* Value before Pruning	*q* Value before Pruning	*P* Value after Pruning	*q* Value after Pruning	Significant LD Tests *a*/*b*[c]
1	IL-6 signaling pathway*	95	0.00012	0.10	0.00012	0.18	7/0
2	Formation and Maturation of mRNA transcript*	172/170	0.00024	0.10	0.00048	0.18	1/3
3	Malaria*	48/46	0.00045	0.10	0.00071	0.18	1/0
4	G13 signaling pathway	39/29	0.01104	0.19	0.00072	0.18	—
5	Cytokine–cytokine receptor interaction*	239/220	0.00126	0.13	0.00136	0.20	24/18
6	Signaling by BMP*	23/17	0.01518	0.20	0.00176	0.20	—
7	Phenylalanine metabolism	17/17	0.00298	0.14	0.00249	0.20	—
8	Pathogenic *Escherichia coli* infection*	52/50	0.00564	0.15	0.00250	0.20	11/2
9	Glycosphingolipid biosynthesis -ganglio series*	14/14	0.00318	0.14	0.00266	0.20	—
10	Advanced glycosylation endproduct receptor signaling	13/11	0.00576	0.15	0.00299	0.20	—
11	Fatty Acid Beta Oxidation*	33/33	0.00416	0.14	0.00316	0.20	—
12	E-cadherin signaling in the nascent adherens junction*	33/23	0.00881	0.17	0.00358	0.20	1/1
13	Visual signal transduction: Rods*	21/21	0.00489	0.15	0.00379	0.20	1/0
14	Regulation of RAC1 activity	38/30	0.00261	0.14	0.00386	0.20	1/3

[a]Gene sets marked with * show a global shift in the distribution of *z* scores, whereas the significance of the others is due to a single high scoring gene.

[b]Previous reports of the involvement of (some genes from) a given pathway in immune response, labeled 1–14 as in the first column: 1, Kishimoto (2010); 2, see supplementary table S8, Supplementary Material online; 3, Barreiro and Quintana-Murci (2010); Hedrick (2011); Fumagalli et al. (2012); 4, Wettschureck and Offermanns (2005); Herroeder et al. (2009); 5, Janeway et al. (2001); 6, Armitage et al. (2011); Dabydeen and Meneses (2011); Portugal et al. (2011); Liu et al. (2012); 7, Boulland et al. (2007); 8, Shaw et al. (2005); 9, Hennet et al. (1998); Bi and Baum (2009); Varki (2009); 10, Harris and Andersson (2004); Bierhaus et al. (2005); Lotze and Tracey (2005); Vasta (2009); 11, Pearce et al. (2009); van der Meer-Janssen et al. (2009); Heaton and Randall (2010); Shriver and Manchester (2012); 12, Lecuit et al. (2000); Cossart and Sansonetti (2004); Nawijn et al. (2011); Van den Bossche et al. (2012); 14, Fischer et al. (1998); Criss et al. (2001); Bokoch (2005); Hebeis et al. (2005); Tybulewicz (2005); Vigorito et al. (2005); Rudrabhatla et al. (2006).

[c]*a* is the number of SNP pairs with *q* values between 10% and 20%; *b* is the number of SNP pairs with *q* values $\leq$ 10%. See supplementary tables S1 and S7, Supplementary Material online, for details about the significant LD links.

an enrichment map (Merico et al. 2010) connects gene sets with at least 33% similarity. This enrichment map includes a large cluster (A) containing 36 gene sets, many of them related to immune response and host defense functions, such as the *Interleukin-6* (*IL-6*) *Signaling* pathway, *Malaria*, and *Cytokine–cytokine receptor interaction*. Interestingly, 53 genes belonging to this cluster A are part of a group of 183 immunity-related genes previously detected by at least two genome wide scans for recent positive selection (Barreiro and Quintana-Murci 2010) (supplementary table S3, Supplementary Material online). A second cluster (B) includes seven pathways, five of which are specifically involved in mRNA processing, such as *Formation and Maturation of mRNA transcript*. These pathways share various genes with the *Influenza Viral RNA Transcription and Replication* pathway, which suggests that the signal of adaptation found in this cluster might be related to host responses to viral infection. A third cluster of similar gene sets (C) contains three pathways related to fatty acid metabolism, such as *Fatty Acid Beta Oxidation* as well as three pathways involved in the metabolism of the amino acids beta-alanine, lysine, and tryptophan. Note that by using the 95% quantile SNP per gene instead of the top scoring SNP, 56 gene sets out of the 70 listed in supplementary table S1, Supplementary Material online, would still be significant, showing that the choice of a non-top scoring SNP per gene leads to broadly comparable results.

To remove the overlap between gene sets, we applied a pruning method inspired by the topGO (Alexa et al. 2006) approach. In short, we started with the most significant gene set and removed its genes from all other sets, and tested again the remaining gene sets. We repeated this procedure with the next most significant gene set until no genes set with more than 10 genes were left. In this way, we end up with a list of pruned pathways that have no overlapping genes and can thus be considered as containing independent information. However, the tests of individual pathways are not independent anymore, and thus the false discovery rate (FDR) needed to be estimated empirically with a permutation approach. Table 1 lists the 14 most significant independent candidate gene sets, which are those sets that score a *q* value equal to or less than 20% both before and after pruning. Interestingly, six gene sets from cluster A were still significant after the removal of shared genes (supplementary fig. S3, Supplementary Material online). It is worth noting that even though we focused only on the 14 most significant candidate pathways in the remaining analyses, the partially overlapping pathways that were lost after pruning might still be of interest and require further investigation.

The Significance of Most Gene Sets Is Not due to Strong Adaptation Signals in a Few Genes but to Small Effects in Many Genes

The 14 significant gene sets can be distinguished into two groups on the basis of their associated *z* score distributions (fig. 2). A first group of four gene sets (*G13 signaling pathway*, *Phenylanaline metabolism*, *Advanced glycosylation end product receptor signaling*, and *Regulation of RAC1* activity) have high scores in the SUMSTAT enrichment test mainly because

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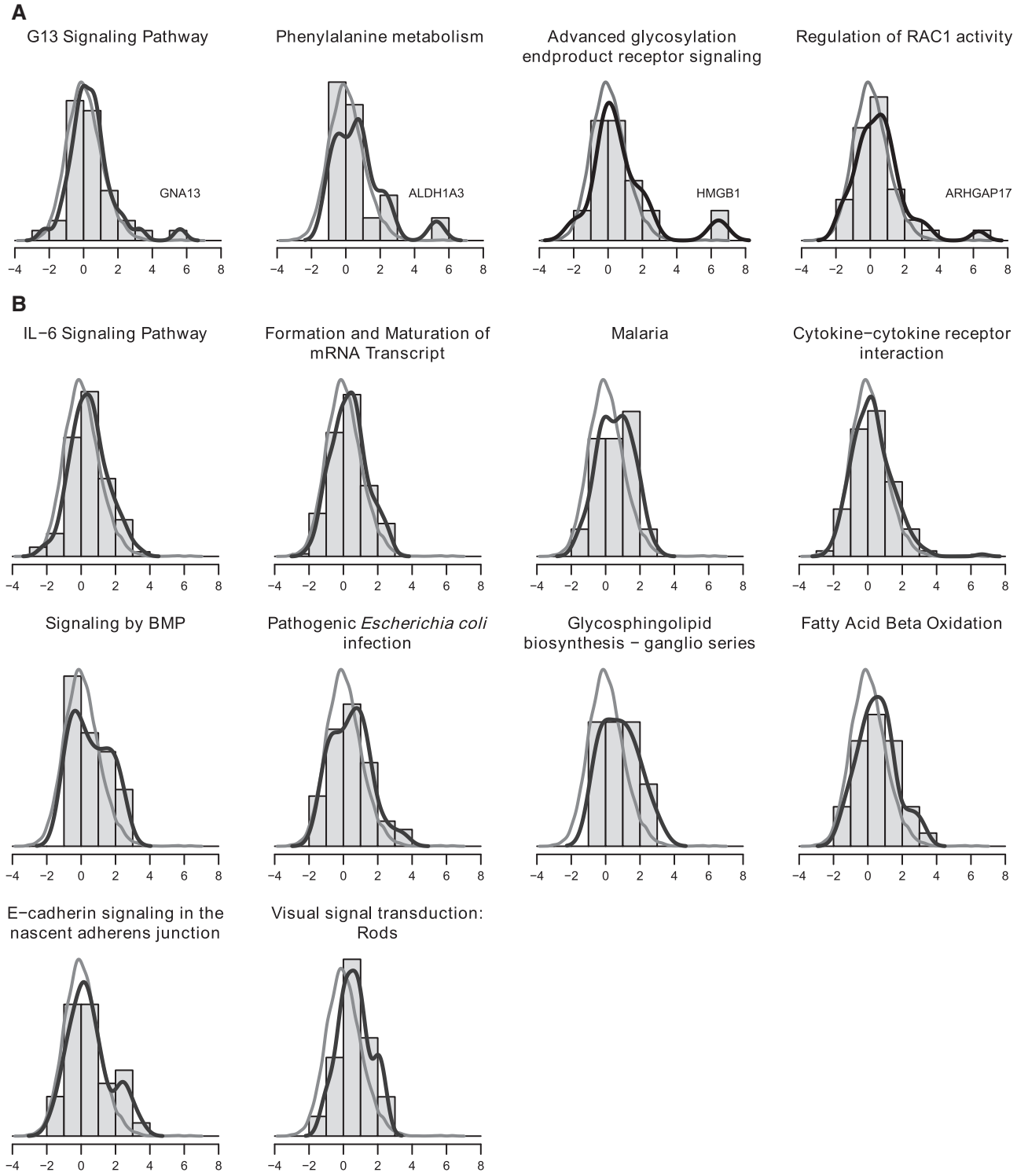


Fig. 2. Distribution of z scores in candidate pathways. These pathways score high in the SUMSTAT enrichment test, because (*A*) they contain a gene with an extreme high z score or (*B*) show a global shift towards large positive z scores. Density plot and histogram of the z scores in the pathway (black line and gray bars) are compared with z scores of all genes (gray line). The names of the extreme scoring genes are reported above the most right bar in (*A*).

they contain one gene (*GNA13*, *ALDH1A3*, *HMGB1*, and *ARHGAP17*, respectively) with a highly significant F_{ST} resulting in a z score larger than 4 (fig. 2A). Without these particular genes, their SUMSTAT score before pruning results in a *q* value higher than the significance threshold of 20%.

On the other hand, the 10 remaining candidate pathways still score a *q* value $\leq 20\%$ after the removal of extreme scoring genes (z score > 4) or the removal of the most extreme gene (supplementary table S4, Supplementary Material online). We thus conclude that these 10 pathways score

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high because the distribution of their z scores is globally shifted to large positive values, implying higher overall levels of population differentiation between populations (fig. 2*B*). The significance of the 10 gene sets in this second group seems therefore due to multiple mutations having gone through incomplete sweeps, rather than to a few mutations with large effects fixed in different populations, which is compatible with moderate levels of positive selection acting on many genes and therefore with polygenic selection. In the remainder of the discussion, we will focus on these 10 candidate gene sets showing signs of polygenic selection.

It is interesting to note that out of the 100 genes with the highest z scores, only 14 genes are present among our 14 candidate gene sets, showing that our pathways are not particularly enriched for outlier F_{ST}s. Furthermore, a commonly used GO enrichment test (with the web tool Fatigo [Al-Shahrour et al. 2004]) on these 100 genes with most extreme F_{ST}s did not reveal any significant biological process. This shows that one taps into a very different type of information when performing gene-set enrichment analysis on all genes as compared with GO enrichment in top scoring genes. Indeed, the conventional GO enrichment approach asks whether the most differentiated loci are overrepresented in certain pathways or GO terms, whereas our enrichment approach addresses the question of which pathways as a whole are most differentiated, which seems more relevant for the detection of polygenic selection.

The Effect of Clustering of Genes in Pathways and Low Recombination Rates

Because recombination rates are negatively correlated with F_{ST} (e.g. Keinan and Reich 2010), a high SUMSTAT score for a given pathway could be obtained if its genes were located in low recombination genomic regions. However, we find that the average recombination rate of the genes in each of the 10 candidate pathways is not significantly lower than in random sets of the same size ($n = 10{,}000$ permutations, $P > 0.05$ for all gene sets), suggesting that the significance of our pathways is not due to low associated recombination rates.

We have also checked whether a possible clustering of functionally related genes of a given pathway could affect our results. Indeed, a single selective event could potentially influence several genes tightly linked on a chromosome, leading to an inflated SUMSTAT statistic and mimicking polygenic selection. To address this issue, we have identified all genes belonging to blocks of 1 cM in length, and we replaced them by a fictive gene with z score computed as the block average. We then recalculated the SUMSTAT score of the reduced pathway and inferred its *P* value as before. As shown in supplementary table S5, Supplementary Material online, all gene sets but one are still found significant before pruning, and sometimes get ranked even higher than with the original approach, which suggests that our results are globally not due to the presence of linked high scoring genes in our pathways. The exception is the *Pathogenic Escherichia coli infection* pathway, which has a new *P* value approximately 10 times larger than in the original analyses, and a new *q* value of 22%, which is slightly above our threshold of 20%. By looking more closely at this latter pathway, we find five regions of 1 cM that contain more than 1 gene (supplementary table S6, Supplementary Material online). Interestingly, four of these five regions harbor two or more functionally related genes from the tubulin or the actin-related protein complex, the latter ones showing similarly high z scores. It follows that the evidence of polygenic selection in the *Pathogenic E. coli infection* pathway could be partly due to the linkage of functionally related genes, even though one cannot exclude that several independent episodes of selection have acted within each 1 cM block.

Signals of Epistatic Interactions within Gene Sets

To test for the presence of potential functional epistasis among loci under selection, we next performed a test of LD at the genotype level (see Materials and Methods) between all genes in our candidate gene sets, where each gene was represented by its associated top-scoring SNP. We first calculated, for each population, the probabilities of two-locus genotype frequencies given the genotype frequencies of each locus, which do not depend on any unknown allele frequencies (Weir 1996). These probabilities were multiplied over populations, and significance was obtained after building a null distribution created by permuting the genotypes at one locus between individuals in a population. Note that this approach respects the underlying genetic structure of the populations, and differs from conventional LD as it does not look for association between specific alleles at different loci, but rather for association between single-locus genotypes.

Interestingly, 7 of the 10 candidate pathways contained genes displaying long distance LD between pairs of top scoring SNPs (q value < 0.2) (table 1). Immune response related pathways presented the strongest evidence of long distance LD (table 1). The *Cytokine–cytokine receptor interaction* pathway showed the largest number of significant scoring pairs (42 pairs of loci with q values < 0.2, fig. 3), followed by *Pathogenic E. coli infection* (13 pairs with q values < 0.2) and the *IL-6 Signaling* pathway (7 pairs with q values < 0.2) (supplementary fig. S4 and table S7, Supplementary Material online). We have tested if our top two pathways showed an excess of significant links with q values $< 20\%$ by creating random gene sets of the same size, testing their top-scoring SNPs for long-distance genotype LD and counting the number of significant links. As this procedure is rather computationally demanding, we only repeated it 100 times. As a result we found only one random pathway with more than 13 connections with q value < 0.2 for the set size of *Pathogenic E. coli infection*, and no random pathways with more than 42 connections with q value < 0.2 for *Cytokine–cytokine receptor interaction*. A possible concern could be that the high number of significant long-distance LD tests is partly caused by genes in short-range LD sharing long-distance LD links. However, this is not the case, as none of the physically clustered genes (less than 500 kb or 1 cM apart) in these gene sets share significant long-distance LD links. We can therefore

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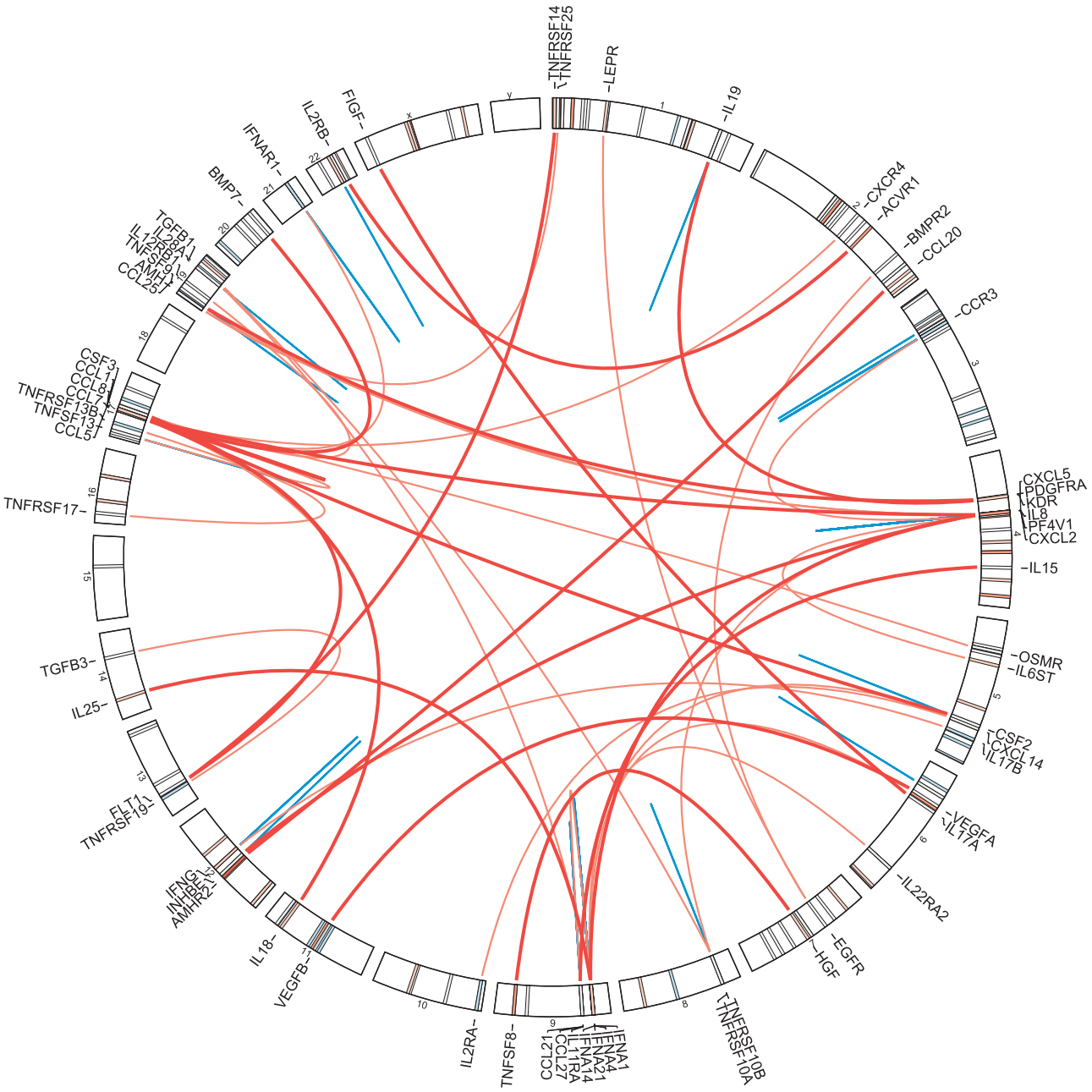


Fig. 3. Long distance genotypic LD in the *Cytokine–cytokine receptor interaction* pathway. All genes in this set are marked on the chromosomes with a color intensity scale corresponding to their standardized z scores (blue, white, red stripes correspond to z scores less than, equal to, or more than zero, respectively). Lines connecting genes correspond to significant genotypic LD (red thick lines: q value $\leq$ 10%, orange thin lines: q value $\leq$ 20%) between the SNPs assigned to these genes. Only genes involved in low q value links are labeled with their gene symbol. Short distance LD, represented by significant links (q value $\leq$ 20%) between SNPs <500 kb apart, is shown in blue.

consider that these two pathways present a significant excess of long-distance LD connections, which could represent signs of epistatic interactions. It suggests that several genes in these pathways have not only evolved adaptively, but have done it in a coordinated manner.

Widespread Signals of Polygenic Selection in Immune Response-Related Pathways

We find a majority of immune response-related pathways among the top candidates for adaptation (table 1). The *Cytokine–cytokine receptor interaction* pathway, which is directly involved in host defense, is particularly interesting since cytokines and their receptors are key regulators of cells engaged in innate and adaptive immune responses (Janeway et al. 2001). Among the various loci displaying evidence of long distance LD in this pathway, the interferon (IFN) family is well represented. IFNs are cytokines that play a key role in innate and adaptive immune responses, and are released by host cells in response to the presence of pathogens or in tumor cells. Among LD-connected genes, it is interesting to note the presence of the type II *IFNG*, whose main function is to trigger anti-mycobacterial immunity, and of

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various genes involved in anti-viral signaling responses, such as members of the type-I IFN family (*IFNA1*, *IFNA4*, *IFNA14*, and *IFNA21*), the first subunit of their common receptor (*IFNAR1*), as well as the type III IFN *IL28A*. These observations overall suggest that IFN responses have evolved in a highly adaptive, and possibly coordinated, manner, which highlights the evolutionary importance of this innate immunity component of host defense (Manry et al. 2011).

Our top scoring *IL-6 Signaling* pathway is an obvious immune defense related pathway as it describes the downstream signaling processes of the cytokine IL-6, which is secreted by T-cells and macrophages and which both stimulate immunoglobulin production by B-cells and regulates T-cell differentiation (Kishimoto 2010).

The *Malaria* and *Pathogenic E. coli infection* gene sets are two other pathways clearly involved in defense against pathogens. Several of the *Malaria* pathway genes are classical examples of loci under selection, including *DARC*, *CR1*, *IFNG*, *CD40LG*, *CD36*, *ICAM1*, *HBB*, *HBA1*, *TNF* (reviewed in Barreiro and Quintana-Murci 2010; Hedrick 2011), and more recently *SELP* (Fumagalli et al. 2012), which shows that our enrichment test successfully discovers pathways that contain several genes directly involved in adaptations. Interestingly, we find quite a large number of significant LD links between SNPs assigned to genes in the *Pathogenic E. coli infection* pathway, of which *TUBA1A*–*TUBB3* and *TUBB2A*–*TUBA3C* score highest (q value $< 10\%$, see supplementary table S7 and fig. S4*D*, Supplementary Material online). These four genes all encode for tubulin subunits, the building blocks of microtubules that are key components of the cytoskeleton and responsible for cell shape and movements. During infection, *E. coli* proteins directly interact with tubulins to disrupt the microtubule structure in the host cell (Shaw et al. 2005), and other bacteria are also able to use host microtubules for invasion (Yoshida and Sasakawa 2003). These results thus suggest that certain combinations of tubulin alleles could be protective against bacterial infection.

Overall, our results show that pathogen-driven selection has been common in the human genome, in agreement with previous observations (Barreiro and Quintana-Murci 2010; Fumagalli et al. 2011), but most importantly that such selective pressures exerted by pathogens have induced polygenic adaptive selection in their human host (Pritchard et al. 2010). Major adaptive episodes could have occurred after the rise of agriculture 10,000 years ago, which might have facilitated the spread of infectious diseases among populations (Barreiro and Quintana-Murci 2010). Different pathogenic environments between regions (Smith and Guegan 2010) could also have resulted in local adaptations in host defense systems.

Polygenic Selection Is Also Observed in Non–Immune-Related Pathways

To a lesser extent than immune-related pathways, other gene sets presented significant evidence of polygenic adaptation (table 1). In some cases, these pathways are also somehow related to host defense, though in an indirect manner. For example, several genes of the *Formation and Maturation of mRNA Transcript* pathway could be involved in viral replication, as 80 genes out of the 172 genes in the pathway are associated with the "Viral reproduction" GO term. Moreover, many of the genes with high z scores in this pathway have been shown to be associated to viral infections (supplementary table S8, Supplementary Material online).

Glycosphingolipids include the ABO and Lewis blood group antigens (Varki 2009), which are associated with protection against several infectious diseases (Anstee 2010), and glycolipids are also used by a variety of viruses and bacteria for cell adhesion and invasion (Varki 2009).

Genes in the *E-cadherin signaling in the nascent adherens junction* pathway are also linked to immune response in various ways. For instance, E-cadherin controls proinflammatory epithelial activity by regulating innate immune functions (Nawijn et al. 2011), it is expressed in a variety of leukocytes (Van den Bossche et al. 2012), and it can be used by bacterial proteins to attach to the host cell and induce cytoskeleton remodeling and plasma membrane extensions necessary for entering host cells (Lecuit et al. 2000; Cossart and Sansonetti 2004).

The *Fatty Acid Beta Oxidation* pathway could have been under selection due to changes in diet or in energy production, but fatty acid oxidation also plays a role in immunity: memory T-cells switch from glucose to fatty acids as energy source (Pearce et al. 2009); the disruption of fatty acid beta oxidation reduces inflammation in the central nervous system (Shriver and Manchester 2012); and viruses can change the lipid metabolism of the host for their own survival (van der Meer-Janssen et al. 2009).

Bone morphogenetic proteins (BMPs), members of the *BMP signaling* pathway, are known for their role in the development of bone and cartilage (Bragdon et al. 2011), and could have been involved in morphological adaptations of human populations in different environments (Ruff 2002). However, stimulation of genes in the *BMP signaling* pathway has been shown to reduce viral infections (Dabydeen and Meneses 2011; Liu et al. 2012) and BMP proteins also regulate iron intake, a potentially important process in infection (Armitage et al. 2011; Portugal et al. 2011).

The *Visual signal transduction: Rods* pathway could have been more specifically affected by environmental adaptations. Rod cells are indeed used in peripheral and night vision because they function at low light levels (Sung and Chuang 2010), and populations living in different environments (e.g., dense forests, and deserts) or extreme latitudes could have developed specific visual abilities.

Conclusions

Until recently, the search for evidence of adaptive evolution in humans has mainly focused on single mutations or on haplotypes restricted to small genomic regions (Nielsen et al. 2005; Voight et al. 2006; Wang et al. 2006; Williamson et al. 2007). However, very few examples of classical selective sweeps induced by positive selection have been found so far (Hernandez et al. 2011), which suggests that human genomic diversity might not have been strongly shaped by positive selection (Lohmueller et al. 2011; Alves et al. 2012), or

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that selection on complex phenotypes has been acting in more subtle ways, for instance by acting on many genes at a time and modifying allele frequencies only slightly (Hancock, Alkorta-Aranburu, et al. 2010; Hancock, Witonsky, et al. 2010; Pritchard et al. 2010). This more complex action of selection makes it more difficult to detect signals of adaptation in our genome.

Genomic scans for selection have been recently criticized for creating a narrative around results in order to validate their methods. Pavlidis et al. (2012) indeed showed that one can always tell a story about selection and local adaptation around any gene, even if it is a false positive. One should thus not validate results a posteriori with the argument that they biologically make sense, and indeed this is not what is done here. Unlike previous approaches testing a posteriori for the enrichment of some biological processes among outlier loci (Hancock, Witonsky, et al. 2010; Hancock et al. 2011), pathways and gene sets are used here rather as input to the analyses and their significance is obtained by explicitly controlling for multiple and non-independent tests.

Although our results are compatible with the action of positive selection acting in populations living in diverse environments at potentially different times, we cannot rule out that they stem from background selection acting in conserved regions (Alves et al. 2012) or from a relaxation of selective constraints (Harding et al. 2000), as these two alternative scenarios would also increase global levels of F_{ST}. It is indeed possible that when migrating out of Africa, some populations were freed from certain pathogens and that constraints on immunity pathways were relaxed, but it appears also likely that the colonization of new environments required specific adaptations to new pathogens, climates, and diet, and that the Neolithic transition and sedentarization was associated with an increased pathogenic load that shaped our immune system (Smith and Guegan 2010). Alternatively, the recurrent elimination of mutations that are less protective to pathogens could be seen as a form of background selection that would decrease the effective population size of the populations and lead to higher levels of genetic drift (Charlesworth 2012), and thus to slightly higher levels of F_{ST}, potentially compatible with what is observed in immune-related pathways (fig. 2).

Although it is unlikely that positive selection has acted on all genes belonging to a canonical pathway, our method still finds 10 candidate gene sets where a sufficient number of its members show collective signals of positive selection. It is indeed much more likely that only a subset of the genes in a pathway rather than a whole pathway or gene set has been responding to selection. In this respect, methods able to detect these subsets should be more powerful than our current approach, but they still need to be developed. We note that our method is also conservative in the sense that it would have difficulty in detecting signals of adaptations in pathways with many genes under strong purifying or balancing selection (associated with low F_{ST} values), as these would have a negative impact on our SUMSTAT statistic. The potential lack of power of our approach might thus prevent us from detecting other instances of polygenic selection, which have less effect on individual fitness than response to pathogens. The fact that 9 out of 10 candidate pathways are directly or indirectly involved with immune response might alternatively suggest that defense against pathogens is the main trait under sufficiently strong selection in humans to be shaping whole pathways, and to lead to a polygenic adaptive response.

It remains to be shown whether the signal we observe in these pathways results from a simultaneous and collective response at many genes at the same time or from successive responses against different pathogens in different environments. Both phenomena might be involved, but the presence of long-distance LD between some pairs of genes suggests that evolution selected for co-adapted allelic combinations. In any case, our study shows that one should move from a narrow gene-centric view of evolution, and give more consideration to whole biological processes as a potential target of selection.

Materials and Methods

SNP Data

We downloaded SNP data from the HGDP-CEPH Human Genome Diversity Panel (Cann et al. 2002; Li et al. 2008) from ftp://ftp.cephb.fr/hgdp_supp1 (last accessed August 6, 2012), which consists of 660,918 SNPs genotyped in 1,043 individuals from 53 worldwide populations. The populations were assigned to five major regions: Africa, Eurasia, East Asia, Oceania, and America according to Rosenberg et al. (2002). We excluded the Uygur and Hazara populations because of their potential admixed status between Eurasians and East Asians (Li et al. 2008). From the remaining 51 populations, we only analyzed the 1,002 individuals that belong to the H1048 subset (Rosenberg 2006), which excludes those individuals with atypical or duplicated DNA. We also removed 188 SNPs located on the Y chromosome, on the pseudoautosomal region of the X chromosome or on mitochondrial DNA. Furthermore, we discarded 12 SNPs that have only missing data, 50 SNPs that were monomorphic in all populations and 4 SNPs that were not typed at all in (at least) one population. We converted the SNP positions on the chromosome from NCBI Build 36.3 to Build 37.3 (UCSC hg19) coordinates. We were unable to map 194 SNPs after this conversion process, leaving us with 660,470 SNPs to be used in further analyses.

Test for Selection

Extreme F_{ST} values can point to candidate loci under selection, but testing the absolute value of F_{ST} is misleading, because it is correlated with heterozygosity (Beaumont and Nichols 1996) (i.e., rare alleles are unlikely to show a large extent of population differentiation, but can still show higher than expected F_{ST} levels). To obtain the expected F_{ST} distribution as a function of different levels of heterozygosity, we ran coalescent simulations under a hierarchical island model of population differentiation as described previously (Excoffier, Hofer, et al. 2009; Hofer et al. 2012) using the program Arlequin (Excoffier and Lischer 2010). In this hierarchical model, demes within the same group (continent) are

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assumed to exchange migrants at a higher rate than demes in different groups, reflecting the hierarchical nature of human continental regions. The joint null distribution of F_{ST} and heterozygosity between populations was generated from 100,000 coalescent simulations, allowing us to infer F_{ST} *P* values and quantiles via a modified kernel density estimation based on a Gaussian kernel instead of the Epanechnikov kernel used previously (Excoffier, Hofer, et al. 2009). The fact that the quantile of a given F_{ST} statistic is evaluated for a given heterozygosity level has also the advantage to take care of the potential SNP assignment bias consisting in an excess of common SNPs in Europe, Asia, and Africa in the HGDP SNP panel (Li et al. 2008). The F_{ST} quantiles were then standardized to *z* scores, using the *qnorm* function of the R program (R Development Core Team 2009). We use these *z* scores as selection test statistics. SNPs can thus be assigned a positive or negative *z* score, corresponding to relatively high or low F_{ST} values, respectively.

Gene Data

From the NCBI Entrez Gene website (Maglott et al. 2011), we downloaded the position of 19,668 protein coding human genes that are located on the autosomes and on the X chromosome (http://www.ncbi.nlm.nih.gov/gene, downloaded on January 4, 2012). For 26 genes, we found multiple locations; in those cases we took the outermost start and end position.

Assignment of *z* Scores to Genes

To have one selection test score per gene, we translated the SNP based *z* scores to gene-based scores. We first assigned SNPs to genes as follows: if a SNP is located within the gene transcript the SNP is assigned to this gene. If a SNP is not located within a gene, it is assigned to the closest gene, provided it is located within 50 kb of the SNP. We thus include SNPs outside genes that might be in LD with yet undiscovered polymorphisms inside genes, as well as SNPs in regulatory regions of a gene. Note that the majority of SNPs (>98%) is thus assigned to only one gene. Next, for each gene *g*, we took the highest *z* score among those SNPs assigned to that gene, which we will refer to as $z(g)$. Alternative methods exist where one uses all SNPs (Holden et al. 2008) or the *n*-ranked SNP (Nam et al. 2010), but in these cases it is difficult to infer a proper null distribution. Note however, that there is a positive correlation between the highest and median *z* score of SNPs assigned to a gene ($r = 0.48$ for all genes and $r = 0.47$ considering genes containing more than one SNP, $P < 2.2\text{e-}16$ in both cases), indicating that the top scoring SNP of a gene is a good representative of the general F_{ST} pattern in that gene. Nevertheless, the use of the highest *z* score among SNPs near a gene can induce a bias, because long genes (with many SNPs) are more likely to show SNPs with extreme values and therefore to be tested significant. A previous gene set enrichment analysis without any correction for SNP number or gene length indeed mostly found gene sets that were enriched for large genes (e.g., *Axon guidance* and *Focal adhesion*) (Amato et al. 2009). To correct for this possible bias in SNP density, we assigned each gene to a bin containing all genes with approximately the same number of SNPs (supplementary table S2 and fig. S1, Supplementary Material online). We then standardized the *z* score based on the *z* score distribution of the bin, using the median based modified *z* score z_{st} (Iglewicz and Hoaglin 1993), which is a robust method in the sense that it is less sensitive to outliers than the common *z* score measure, and defined as

$$z_{st}(g) = \frac{0.6745(z(g) - \text{median}(z(g)_{\text{bin}})}{\text{MAD}(z(g))_{\text{bin}}}, \quad (1)$$

with MAD denoting the median absolute deviation computed as

$$\text{MAD}(z(g))_{\text{bin}} = \text{median}_{i,\, g_i \in \text{bin}}\big(\,|\, z(g_i) - \text{median}(z(g))_{\text{bin}}\,|\,\big). \quad (2)$$

Note that the constant 0.6745 is the expected value of MAD for a normal distribution and large sample size, expressed in units of standard deviation. For ease of reading, we will refer to these bin-standardized *z* scores simply as *z* scores in the remaining part of our analyses. We removed 1,750 genes that did not have any SNPs in their direct neighborhood. The remaining 17,918 genes were used as reference list in our enrichment tests, and we shall call this list *G* in our further analyses.

Gene Sets

Currently, many pathway databases are publicly available, such as KEGG (Kanehisa et al. 2012), REACTOME (Matthews et al. 2009), or the Pathway Interaction Database (PID) (Schaefer et al. 2009). The NCBI Biosystems database (Geer et al. 2010) includes pathways from these and other databases, which we use as a source of a large collection of gene sets in a standard format. We downloaded 2,019 human gene sets from the NCBI Biosystems database (Geer et al. 2010) on March 23, 2011 from http://www.ncbi.nlm.nih.gov/biosystems. We removed genes that could not be mapped to the gene list *G*. Furthermore, we discarded gene sets with less than 10 genes, leaving us with 1,149 genes sets. Finally, we identified 75 groups of (nearly) identical gene sets, namely those sets sharing at least 95% of their genes, and replaced these groups by single gene sets ("unions") consisting of all genes in such a group (supplementary table S1, Supplementary Material online). The remaining 1,043 gene sets served as input in our enrichment tests (see supplementary tables S1 and S9, Supplementary Material online, for more information on the properties of gene sets and genes).

Genetic Distance and Recombination Rates

We downloaded local recombination rates and the genetic map coordinates of phase 2 HapMap SNPs on March 5, 2013, from http://hapmap.ncbi.nlm.nih.gov/downloads/recombination/2011-01_phaseII_B37. We could map almost all of our top SNPs assigned to genes to the SNPs in this table. For a few SNPs (180), there was no exact match, and we estimated their local recombination rate and genetic map

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coordinates by a linear interpolation using the two closest SNPs in the HapMap table.

Enrichment Test

To test for enrichment of signals of selection in the gene sets, we calculated the SUMSTAT score (Tintle et al. 2009) for each gene set *S*, which takes simply the sum of the *z* scores of genes in a gene set as,

$$\mathrm{SUMSTAT}(S) = \sum_{g \in S} z_{st}(g) \quad (3)$$

The significance of SUMSTAT(*S*) was assessed by comparing it with a null distribution of SUMSTAT scores of random gene sets *S'* chosen to have the same size as the original set. According to the Central Limit Theorem the SUMSTAT scores of these random gene sets should approach a normal distribution (Rice 2007). Therefore, instead of generating a huge amount of random gene sets to create a null distribution, we inferred the *P* values from a normal distribution, with the mean ($\mu_{S'}$) and the variance ($\sigma^2_{S'}$) computed from the mean (μ_G) and variance (σ^2_G) of $z_{st}(g)$ in gene list *G* (the set of all 17,918 genes to which we can assign SNPs) as $\mu_{S'} = n\mu_G$, and $\sigma^2_{S'} = n\sigma^2_G$, respectively, with *n* being the number of genes in the gene set. Supplementary figure S5, Supplementary Material online shows that SUMSTAT scores of random sets indeed approximate a normal distribution. We used the *pnorm* function from *R* to compute the *P* values assuming this normal distribution.

As we tested a large number of gene sets (1,043), we need to correct for multiple testing. We therefore calculated the *q* value (Storey and Tibshirani 2003) from the *P* values of our tested gene sets. Briefly, the *q* value of a gene set with *P* value *P** is the expected FDR at which all gene sets with a *P* value $\leq P^*$ would be called significant. The *q* value thus includes a FDR correction for multiple tests. We considered gene sets with *q* value ≤ 0.2 to be candidate gene sets for positive selection, thus allowing for 20% false positives among these candidates. We did the calculations using the function *q* value with default parameters from the R package *q* value based on the method developed by Storey and Tibshirani (2003).

To test whether potential candidate gene sets were sensitive to the removal of extreme genes, we removed extreme scoring genes (genes with *z* score > 4, a clear group of outliers in the distribution of *z* scores of all genes) from all 1,043 pathways and recalculated their SUMSTAT score. To assess significance, these scores were compared with a null distribution of random gene sets built from the gene list *G* with extreme scoring genes removed. We performed a similar test, but this time with the highest scoring gene removed from the tested sets, irrespective of its *z* score, and we tested the significance of SUMSTAT scores by building a null distribution of random sets with their highest scoring genes removed as well.

Enrichment Map

The enrichment map in figure 1, which shows the similarity between significant pathways after testing gene sets for enrichment of signals of selection, was created with the Cytoscape (Smoot et al. 2011) plug-in *Enrichment Map* (Merico et al. 2010). We set the overlap coefficient cutoff to 33%, the *P* value cutoff to 1.0 and the FDR *Q* value cutoff to 0.2, with the latter meaning the *q* value as described earlier.

Removing Genes from Overlapping Gene Sets

Many gene sets share a considerable amount of genes, and we applied a pruning method inspired by the topGO approach described in (Alexa et al. 2006) to remove any gene redundancy between gene sets. Note that a similar approach has been used by George et al. (2011). In the topGO method, significant genes are removed from parent GO terms when testing for GO term enrichment. In our approach, we used the following steps. With a list *L* of gene sets to be tested and the list *G* of genes:

1) Test all gene sets in *L* and rank the sets on *P* value (from lowest to highest *P* value).
2) Remove the first set *S* from *L* and store it in a new list *L'*.
3) Remove the genes in *S* from the remaining gene sets in *L* and from the gene list *G*.
4) Remove all sets in *L* that are smaller than an arbitrary minimum set size *n* (we used here $n = 10$).
5) If *L* contains more than one set: go back to 1.
6) Rank the sets in *L'* on *P* value and empirically correct for multiple testing (discussed later).

Empirical Correction for Multiple Testing after Pruning the Gene Sets

The remaining gene sets in *L'* are not independent and their *P* values are therefore biased: the *P* values of the sets before pruning are approximately uniformly distributed, while after pruning there is a bias towards low *P* values (supplementary fig. S6, Supplementary Material online). Consequently, we could not apply standard FDR or *q* value calculations, and we used instead a randomization method to estimate FDR and *q* values.

If we reject all hypotheses with a *P* value less than a given threshold *P**, we can estimate the FDR with

$$\hat{FDR}(P^*) = \frac{\pi_0 \cdot \hat{V}(P^*)}{R(P^*)}, \quad (4)$$

where π_0 is the proportion of true null hypotheses, $\hat{V}(P^*)$ is the estimated number of rejected true null hypotheses if all hypotheses are true nulls and $R(P^*)$ is the total number of rejected hypotheses. If the tests are independent the *P* values of the true null hypotheses are uniformly distributed and $\hat{V}(P^*)$ could be estimated with $\hat{V}(P^*) = P^* m$, where *m* is the number of hypotheses (Storey and Tibshirani 2003). However, in our case the hypotheses are not independent. To estimate $\hat{V}(P^*)$, we repeatedly ($n = 200$) permuted the *z* scores in the gene list *G* and tested the gene sets with the pruning method as described earlier. $\hat{V}(p^*)$ was then calculated from the mean proportion of *P* values $\leq P^*$ in the permuted sets. We used a histogram based method to estimate π_0 (Nettleton et al. 2006). In short, this algorithm computes

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the number of true null hypotheses by iteratively comparing the histogram of observed P values with the expected P value frequencies of the true null hypotheses. We describe this method in more detail in supplementary text S1, Supplementary Material online, and illustrate the iteration steps in supplementary figures S7 and S8, Supplementary Material online.

We calculated $\hat{FDR}(P^*)$ for a large range of P values in [0, 1], and we estimated the q value, $q(P^*)$ as the minimum FDR corresponding to any P value greater than or equal to P^*:

$$\hat{q}(P^*) = \min_{P':P' \geq P^*} \left(\hat{FDR}(P')\right). \quad (5)$$

Supplementary figure S9, Supplementary Material online, depicts the FDR and q value estimates for a range of P values. We constructed the list with candidate gene sets by selecting those gene sets that score a maximal q value of 20% before pruning and after pruning.

Testing for Genotypic LD

We collected individual genotypes for all SNPs assigned to genes in each candidate pathway (including those genes that were removed after pruning), and we tested for genotypic LD between pairs of loci using an exact test. For all pairs of SNPs in a set, we created a contingency table per population with the two-locus-genotype counts and marginal single-locus genotype counts. Individuals with missing data in one or two of the SNPs were removed. Assuming independence in the entries of the contingency tables, we estimated the probability of the observed two-locus-genotype counts conditional on the single-locus counts as (Weir 1996):

$$\Pr(n_{ijkl} \mid n_{ij}, n_{kl}) = \frac{\prod_{ij} n_{ij}! \prod_{kl} n_{kl}!}{n! \prod_{ijkl} n_{ijkl}!}. \quad (6)$$

We then calculated the overall probability of LD by taking the product over all populations:

$$\begin{aligned} \Pr_{\text{overall}} &= \prod_{d \in \text{pops}} \Pr(n_{ijkl}(d) \mid n_{ij}(d), n_{kl}(d)) \\ &= \prod_{d \in \text{pops}} \frac{\prod_{ij} n_{ij}(d)! \prod_{kl} n_{kl}(d)!}{n(d)! \prod_{ijkl} n_{ijkl}(d)!} \end{aligned} \quad (7)$$

with pops being the set of populations, and $n(d)$, $n_{ij}(d)$, $n_{kl}(d)$, and $n_{ijkl}(d)$ being the genotype counts in population d. We performed an exact test by repeatedly permuting in each population the genotypes at one locus while keeping the genotypes at the other locus fixed and calculating $\Pr(n_{ijkl} \mid n_{ij}, n_{kl})$. We then compared the observed $\Pr_{\text{overall}}$ with our empirical null distribution to assess its P value. To reduce computation time, we apply a sequential random sampling method (Besag and Clifford 1991), meaning that we stepwise increase the null distribution until it becomes clear that the null hypothesis will never be rejected or that the null distribution has reached a maximum size. A more detailed description of this method can be found in supplementary text S2, Supplementary Material online. Finally, P values were corrected for multiple testing using the function q value with default parameters from the R-package q value; those tests with a q value less than 20% were reported (supplementary table S1, Supplementary Material online). As we found many significant interactions between homologous genes, a concern could be that these are due to misannotation of the probes on the SNP chip. We thus used BLAST (http://blast.ncbi.nlm.nih.gov/, last accessed August 21, 2012) to confirm that the probes on the Illumina HumanHap650Y SNP Beadchip (Li et al. 2008) were mapped to the correct chromosome location. Visualization of the position of genes on chromosomes and significant linkage between them in figure 3 and supplementary figure S4, Supplementary Material online, was done with the program Circos (Krzywinski et al. 2009).

Supplementary Material

Supplementary text S1 and S2, tables S1–S9, and figures S1–S9 are available at *Molecular Biology and Evolution* online (http://www.mbe.oxfordjournals.org/).

Acknowledgments

The authors thank three anonymous reviewers, Luis Barreiro, and Jeff Jensen for their extensive and helpful comments on the manuscript, as well as Julien Roux, Ioannis Xenarios, and Mark Ibberson for discussions and interesting suggestions at different phases of this research. This work was supported by a Swiss National Science Foundation grant (PDFMP3-130309) to L.E.

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